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Brain Tumors: Grade Level Diagnosis Through the Combination of Meta-Heuristic Algorithms and Clustering Protocols in MRI Images

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Abstract:	Glioma brain tumors are the most common type of primary brain tumor, all of which originate from glial cells that surround neurons. Computer-aided diagnosis methods have aided neurologists in diagnosing this disease. Recent advancements have led to improved efficiency with the emergence of deep learning. The approaches of computer-aided recognition systems include preprocessing, segmentation, feature analysis (feature extraction, feature selection, and feature verification), and classification. In this study, image processing, machine learning, and deep learning are used to identify glioma brain tumors and classify them into high-grade and lowgrade types. The pre-processed images are segmented using the K-Means clustering algorithm. At this stage, two meta-heuristic algorithms, Coyote and CoronaVirus, have been used to optimize K-Means. Next, deep features are extracted through convolutional neural networks, and tissue features are extracted using local binary patterns from images related to the tumor area. The dataset used in this research is Brats2018, which includes 210 MRI images of highgrade glioma tumors and 75 images of low-grade glioma tumors. In the first proposed method, image segmentation was done using a combination of the K-Means clustering algorithm and the Coyote optimization algorithm, as well as different feature extraction methods, including texture feature extraction using the local binary patterns method and deep feature extraction. The extraction of deep features from the pre-trained VGG16 network yielded the best results, achieving an accuracy of 99%. In the second method, the clustering centers were optimized using the coronavirus algorithm, which achieved the best performance with an accuracy of 99.80%.
Suggested Reviewers:	Zekuan Yu, PhD Associate Professor, Fudan University yzk@fudan.edu.cn The reviewer's recent research "HSA-net with a novel CAD pipeline boosts both clinical brain tumor MR image classification and segmentation" is directly relevant to the present research work and has been cited in the manuscript. Their insights could be valuable in evaluating the context and contribution of the research.
	M. Jamal Deen, PhD Professor, McMaster University jamal@mcmaster.ca The recent studies conducted by this reviewer such as "Brain tumor segmentation and grading of lower-grade glioma using deep learning in MRI images" align closely with the present research and have been referenced in the manuscript. Their expert perspective could be instrumental in assessing the relevance and impact of the work.

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To: Editor-in-chief of **Artificial Intelligence in Medicine**,

Dear Dr. Carlo Combi,

I am so delighted to submit the manuscript entitled "**Brain Tumors: Grade Level Diagnosis Through the Combination of Meta-Heuristic Algorithms and Clustering Protocols in MRI Images**" for consideration for publication in **Artificial Intelligence in Medicine**. The research addresses the critical need for accurate diagnosis of glioma brain tumors, through a novel computer-aided diagnosis method. The proposed methodology incorporates the latest image processing, machine learning, and deep learning techniques to distinguish between high-grade and low-grade glioma tumors with remarkable accuracy.

The core of the proposed methodology is the novel use of the K-Means clustering algorithm, optimized by Coyote and CoronaVirus meta-heuristic algorithms, to segment brain MRI images. According to findings, on the Brats2018 dataset comprising 210 high-grade and 75 low-grade glioma MRI images, the proposed methodology demonstrates superiority over state of the art. The first method, combining K-Means clustering with Coyote optimization and various feature extraction techniques, achieved an accuracy of 99%. The second method, optimized by the coronavirus algorithm, could achieve an accuracy of 99.80%.

We are confident that the manuscript will be of interest to the readership of Artificial Intelligence in Medicine due to its contribution to the Medicine field and its potential to significantly enhance the glioma tumor diagnosis.

The authors have no conflicts of interest to declare and have approved the manuscript and agree with its submission to Artificial Intelligence in Medicine. Enclosed with this letter are the highlights and the manuscript for your review. I confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication by another journal.

Thank you for your time and consideration. We look forward to hearing your positive response.

Yours sincerely,

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Highlights

- Employing computer-aided diagnosis methods to classify glioma brain tumors.
- Integrating Coyote and CoronaVirus meta-heuristic algorithms for image segmentation.
- Ensemble and Particle Optimization Algorithm-Based feature extraction and selection.
- Coyote-optimized K-Means clustering reached 99% accuracy, using the VGG16 features.
- Further optimization of K-Means by Coronavirus led to a higher accuracy of 99.80%.

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ABSTRACT

Glioma brain tumors are the most common type of primary brain tumor, all of which originate from glial cells that surround neurons. Computer-aided diagnosis methods have aided neurologists in diagnosing this disease. Recent advancements have led to improved efficiency with the emergence of deep learning. The approaches of computer-aided recognition systems include preprocessing, segmentation, feature analysis (feature extraction, feature selection, and feature verification), and classification. In this study, image processing, machine learning, and deep learning are used to identify glioma brain tumors and classify them into high-grade and low-grade types. The pre-processed images are segmented using the K-Means clustering algorithm. At this stage, two meta-heuristic algorithms, Coyote and CoronaVirus, have been used to optimize K-Means. Next, deep features are extracted through convolutional neural networks, and tissue features are extracted using local binary patterns from images related to the tumor area. The dataset used in this research is Brats2018, which includes 210 MRI images of high-grade glioma tumors and 75 images of low-grade glioma tumors. In the first proposed method, image segmentation was done using a combination of the K-Means clustering algorithm and the Coyote optimization algorithm, as well as different feature extraction methods, including texture feature extraction using the local binary patterns method and deep feature extraction. The extraction of deep features from the pre-trained VGG16 network yielded the best results, achieving an accuracy of 99%. In the second method, the clustering centers were optimized using the coronavirus algorithm, which achieved the best performance with an accuracy of 99.80%.

1. Introduction

Today, with the increase in the prevalence of cancer, including brain tumors, and the decrease in the age of its onset, engineers and doctors are seeking suitable solutions for timely diagnosis and treatment of such diseases. Brain tumors, being one of the most common causes of human mortality, make their timely diagnosis particularly important. Cancer can be defined as the abnormal growth and division of body cells. A brain tumor, which is a type of cancer, occurs in the brain itself or in the nerves of the skull, meninges, and pituitary gland (Işın, Direkoğlu and Şah, 2016). Medical imaging encompasses various methods that allow the identification of the smallest abnormalities in the human body. The primary goal of medical imaging is to extract meaningful and accurate information from these images with the least possible error (Kapoor and Thakur, 2017). In this regard, different imaging techniques such as computed tomography, positron emission tomography, and magnetic resonance imaging are used to diagnose brain tumors. In the MRI imaging process, no drugs are injected into the body, there is no radiation damage, and the whole process is completely safe (Ilhan and Ilhan, 2017).

Different patterns of indentations and grooves in the brain, as well as noise in medical images, have posed challenges in diagnosing these types of diseases. Meanwhile, computer methods and algorithms based on feature extraction can more quickly identify the tumor location. These methods aid in the analysis of tumors in MRI images. Computer-based models rely more on MRI images because they provide more contrast for soft tissues compared to CT images. These models include stages of pre-processing, segmentation, feature analysis (feature extraction, feature selection, and feature verification), and classification.

The initial stage, known as pre-processing, aims to enhance the quality of medical images. Following this, brain tumor segmentation is performed to divide the image into homogeneous regions, aiding in the determination of the regions shape. Grading brain tumors is a significant challenge due to their variety in terms of shape, size, and associated

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cognitive impairments. The objective of this research is to diagnose glioma brain tumors and classify them into high and low grades using image processing techniques, machine learning algorithms, and deep learning.

Accurate disease diagnosis by doctors necessitates the extraction of correct and reliable information from available images and data. Tumor identification research encompasses both tumor detection and grading. The former aims solely at disease identification and diagnosis, while the latter seeks to determine the diseases severity in addition to its diagnosis. The design of automatic methods for disease diagnosis is of great importance today. Recent studies have shown that the use of machine learning and deep learning algorithms has produced satisfactory results in the identification of brain tumors.

1.1. Research objectives

This research aims to segment brain cell tissues using machine learning algorithms and extract crucial features using tissue features and deep features, providing modern approaches for brain tumor identification. Specifically, this research seeks to answer the following questions:

- Can we diagnose tumor tissues using clustering algorithms and meta-heuristic algorithms?
- Does feature extraction from the tumor area, using the LBP method and convolutional neural networks, provide the best image features?
- Does the proposed method outperform traditional methods in classifying glioma into high-grade and low-grade types?

This research is organized into five sections. Section 2 outlines the fundamentals of the research aligned with the implementation of the proposed method. Section 3 provides a detailed description of the proposed methods. Section 4 presents the implementation details, and Section 5 showcases the results obtained from the proposed methods for diagnosing and grading brain tumors. Finally, Section 6 concludes the results and proposes directions for future research.

2. Related Work

Digital image processing, a burgeoning technology, has diverse applications in fields such as medicine, military and security sciences, geology, astronautics, astronomy, and urban planning. These digital images, displayed with a limited number of bits, are arrays of real or mixed numbers. Thus, images obtained from imaging tools are first digitized and then stored in computer memory as binary numbers (Mohan and Subashini, 2018). Biomedical image processing, a field that combines biomedical engineering and computer science, has become one of the most effective techniques for monitoring health status. Advances in this field allow for the detection of masses, whether they are in internal organs or not, using image processing techniques. Systems employing this method are known as Computer-Aided Diagnosis (CAD) systems. The goal of these systems is to enhance the accuracy of tumor diagnosis and assist radiologists. CAD systems encompass preprocessing, segmentation, feature analysis, and classification (Ulku and Camurcu, 2013).

In medical image processing, the quality of images used in diagnostic systems is a crucial factor that directly influences the final success rate. Therefore, the first stage is pre-processing, aimed at enhancing the quality of medical images (Ulku and Camurcu, 2013). Preprocessing involves processes such as converting the image to grayscale, reducing and removing noise, reconstructing the image, and in the case of medical images, removing hard tissues like the skull from MRI (Kapoor and Thakur, 2017). MRI images typically contain noise such as Gaussian, salt and pepper, and speckle noise. Intensity normalization is another critical step in MRI preprocessing, particularly when classification and clustering methods are employed for segmentation. However, segmenting tumor images is more challenging than healthy images due to confounding effects caused by variations in brain tumor appearance (Liu, Li, Wang, Wu, Liu and Pan, 2014).

Segmentation divides an image into its constituent elements. The input for this step is an image, and the output is the features extracted from that image. This process facilitates further analysis of the images and extraction of meaningful information, leading to the sharing of pixels in a common feature (Kapoor and Thakur, 2017). In the medical field, various segmentation methods divide the MRI brain into white matter, gray matter, and cerebrospinal fluid. These

methods, which include manual, semi-automatic, and fully automatic segmentation, are classified based on the level of human interaction required. They can also be grouped based on the features used, such as the threshold-based, region-based approaches, supervised, unsupervised segmentation, and segmentation based on neural networks (El-Dahshan, Mohsen, Revett and Salem, 2014).

Accordingly, feature extraction subsequently identifies and isolates the essential components of digital images, simplifying complex input data into a set of reduced features. Features are generally categorized into two broad types: general (such as color, shape, and texture) and domain-specific (conceptual features and human face). Among the latest feature extraction methods are those based on deep learning, which are utilized in various problems such as object recognition, image classification, and image segmentation (Phung and Rhee, 2019). Convolutional neural networks, a type of artificial neural network that mimics human vision, fall under this category. These networks comprise different layers, including the convolutional layer, integration layer, and fully connected layer, the general form of which is depicted in Figure 1.

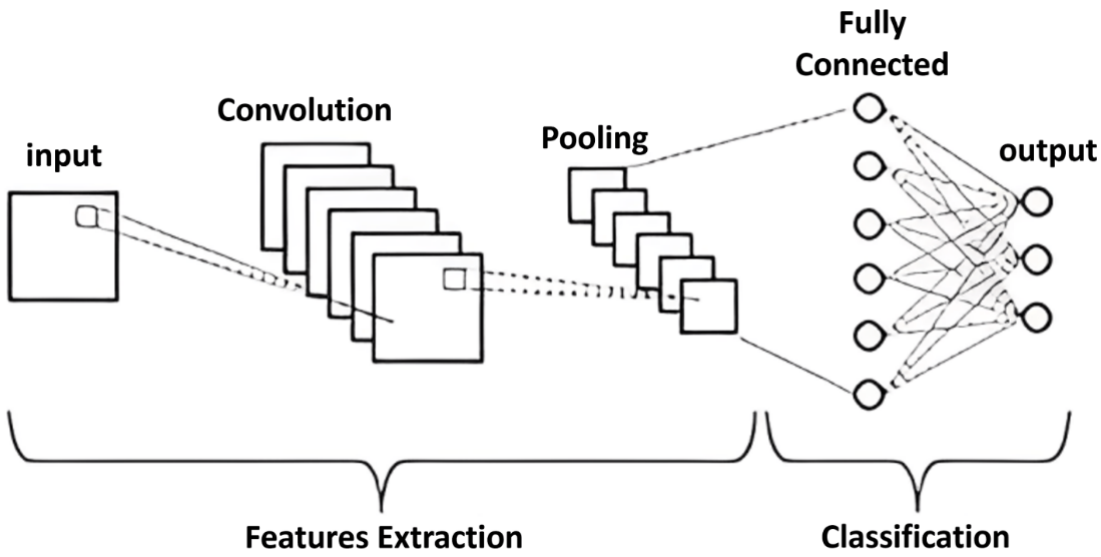

Figure 1: Convolutional neural network architecture (Albawi et al., 2017).

On the other hand, there are pre-trained neural networks that are trained on large datasets and can apply their learned knowledge to various problems. This research utilizes the pre-trained VGG16 network, which we will discuss further in the next section. The number 16 in this model refers to the number of network layers. This network, like the basic CNN model, consists of three main layers that were mentioned earlier. The VGG16 network is trained on the ImageNet dataset, which contains half a million images for training, 50,000 images for validation, and 100,000 images for testing (Vrbancic, Zorman and Podgorelec, 2019). ImageNet comprises 1000 classes or labels. The architecture of the VGG16 model is depicted in Figure 2.

In this context, the analysis of high-dimensional data poses various challenges for professionals in the fields of machine learning and data mining. Feature selection plays a crucial role in reducing processing time and improving accuracy by removing irrelevant and redundant features. By selecting a specific subset of features according to certain criteria, the workload of data processing is significantly reduced. Feature selection can be categorized into supervised and unsupervised methods. Supervised feature selection focuses on the relationship between features and class labels, leading to enhanced classification model performance, reduced overfitting, and lower computational training costs. On the other hand, unsupervised feature selection methods aim to enhance clustering by identifying a subset of features based on clustering or evaluation criteria (Cai, Luo, Wang and Yang, 2018).

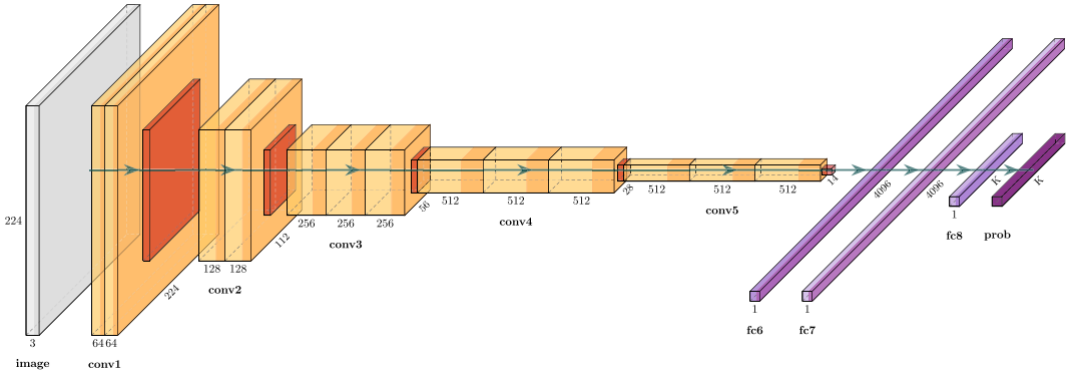

Figure 2: VGG16 architecture (Vrbancic et al., 2019).

There are multiple approaches to achieve feature selection, including filter-based methods, wrapper methods, and embedded methods. Filter-based methods evaluate the significance of each feature individually, disregarding the influence of other features, and then rank them using statistical criteria such as correlation or mutual information. These methods are efficient in terms of computation and can be applied to both supervised and unsupervised classification tasks. Wrapper methods evaluate subsets of features based on their performance quality on a modeling algorithm, and the predictive performance of the model is used to assess the usefulness of a particular subset of features. These methods outperform filter-based methods because the selected feature set is evaluated using an actual classification model. Embedded methods perform feature selection as part of the model training process. These methods typically work with classifiers such as decision trees and logistic regression and consider penalties for features that do not contribute to the model (Jović, Brkić and Bogunović, 2015).

Various studies have identified brain tumors from MRI images. One such study (Sharif, Amin, Raza, Yasmin and Satapathy, 2020) identified brain tumor regions based on the particle swarm algorithm. The proposed method preprocesses to enhance medical images and remove non-brain regions from MRI images, thereby reducing processing time. Following this, the images are segmented using the particle swarm algorithm, where the intensity levels of the pixels are considered as potential solutions. Subsequently, local binary patterns and deep features are extracted from the segmented images using capsule neural networks. A genetic algorithm is then utilized to choose the most optimal feature. Finally, the artificial neural network and other classification models are employed to detect the severity of the tumor. The process of segmenting and extracting the tumor area in this study is depicted in Figure 3.

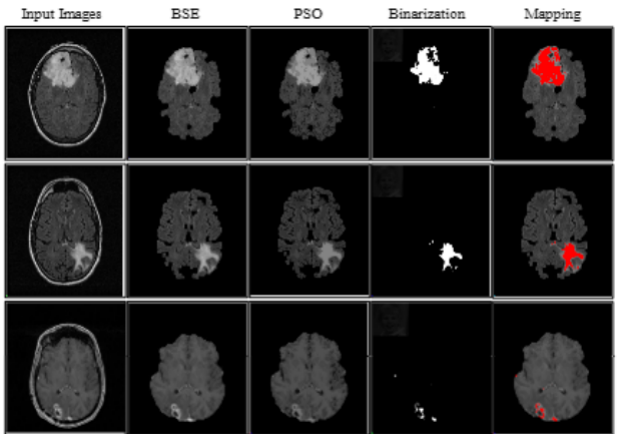

Figure 3: Segmentation and tumor region extraction steps (Sharif et al., 2020).

In the method proposed by the researchers (Tousan, Mehrshad and Ghaemi, 2016), the tumor position is identified using a threshold-based method. Subsequently, this position is used as an indicator in a basin segmentation method. The results of this study demonstrate that the proposed method for precise detection of the tumor area yields consistent and accurate results across a wide range of variations in the input parameters. The method proposed in this study is illustrated in Figure 4.

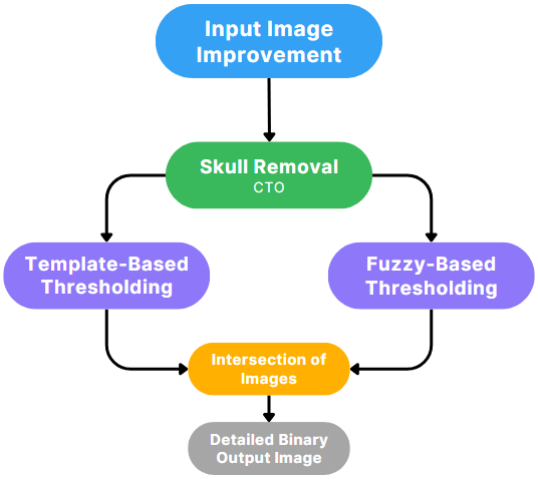

Figure 4: Proposed methodology in (Tousan et al., 2016).

The study (Saii and Kraitem, 2017) employs the histogram equalization technique to enhance the resolution of Word images. Following this, a segmentation process is applied to extract Regions of Interests (ROIs) or background objects from the images. At this stage, morphological image processing techniques are utilized to identify the ROI. The features obtained are then inputted into the Rapid Miner program in the form of an Excel file for classification. The segmentation method presented in this paper achieved an accuracy of 95.5%.

Another study (Gopal and Karnan, 2010) segments tumor regions using two genetic algorithms and particle optimization. These algorithms automatically determine the optimal threshold value in the provided images for the initial point of the cluster. Subsequently, the Fuzzy C-Means clustering algorithm calculates an adaptive threshold for brain tumor segmentation. In this study, the average accuracy of the genetic algorithm is 89.6%, while the particle optimization method achieves an accuracy of 92.8%. For tumor detection, the accuracy reaches 97.87%.

The method proposed in the study (Amin, Sharif, Raza, Saba, Sial and Shad, 2020) consists of two phases: pre-processing and feature learning. The latter uses LSTM, a type of deep recurrent neural network, along with fully connected and softmax layers for binary classification of MRI images. MRI images may be affected by coil fluctuations in the magnetic field, necessitating the estimation and correction of the bias field. To address this issue, the researchers in (Amin et al., 2020) use the N4ITK preprocessing method to correct the bias field and the Gaussian filter to smooth the image. In the feature extraction stage, a deep learning model consisting of four LSTM layers, a fully connected layer, a softmax layer, and a classification layer is used. This study achieved 98% accuracy using the proposed method and the Brats 2018 dataset.

In another study (Rehman, Khan, Saba, Mehmood, Tariq and Ayesha, 2021), the researchers selected the best set of features extracted from MRI images by the pre-trained VGG19 network using a correlation-based method. These features were evaluated using a feed-forward neural network. This method calculates the correlation between features and labels and selects the best feature set using a greedy search algorithm. The final classification of these features is performed by the fully connected layer, with the output determined by the number of classes in the problem. This

approach achieved an accuracy of 92.67%.

The research (Kapoor and Thakur, 2017) utilizes the region-based watershed segmentation technique for image segmentation tasks. This technique identifies the region to which each pixel belongs by searching for pixel and area similarity. The paper introduces the watershed transformation function, which starts with a morphological operation and then creates an intensity gradient of the input image. Finally, the gradient image is used in conjunction with the watershed transformation function to conduct the final contour search.

Manual segmentation of brain images is a time-consuming and error-prone task, hence the utility of deep learning models. In this context, the researchers of the study (Daimary, Bora, Amitab and Kandar, 2020) utilized U-Net, SegNet, and ResNet18 for automatic image segmentation. Various combinations of these three architectures were tested in this research, with the U-Net-SegNet combination model yielding the best resultan average accuracy value of 99.11%.

The research (Basheera and Satya Sai Ram, 2019) investigates the convolutional neural network for segmentation and classification of medical images. In the proposed method, MRI images are first segmented using independent component analysis. Subsequently, deep features are extracted from the segmented image and classified. The proposed method of this study provides techniques for identifying brain tumors from MRI images with the aim of determining the tumor grade. These techniques include approaches based on machine learning, deep learning, and image processing.

Gliomas, the most prevalent type of malignant brain tumors, have varying grades that significantly impact patient survival rates. To address this, the study by Naser and Deen (2020) developed a deep learning approach. Their approach combines a U-net-based Convolutional Neural Network (CNN) for tumor segmentation and a transfer learning technique using a pre-trained Vgg16 convolutional base and a fully connected classifier for tumor grading.

In a different research conducted by Başaran (2022), a new computer-based hybrid diagnostic model is suggested with a high accuracy rate in diagnosing normal brains and brains with various types of tumors. This is achieved through the utilization of brain images obtained via MRI techniques. Feature selection is performed using genetic algorithms, particle swarm optimization techniques, and artificial bee colony optimization techniques. Metaheuristic optimization algorithms are employed for feature selection, and the unique features of the images are subsequently classified using support vector machine kernels.

Although deep learning models have made great strides in medical segmentation, their capabilities are restricted to generating segmentation maps without accounting for the uncertainty involved in the process. In order to tackle this limitation, the research conducted in (Zhou and Zhu, 2023) incorporated uncertainty quantification into the deep learning model and applied it to the segmentation of brain tumors using multiple modalities. Additionally, they devised an attention-aware multi-modal fusion technique that efficiently captures complementary feature information from various MR modalities.

Afterwards, a highly effective and cutting-edge model was introduced in the study (Emam, Samee, Jamjoom and Houssein, 2023), which utilizes deep learning and enhanced metaheuristic algorithms. The researchers devised a specialized optimized residual learning architecture to accurately classify various brain tumors. Additionally, they introduced an enhanced version of the Hunger Games Search algorithm (I-HGS) by incorporating two strategies: the Local Escaping Operator (LEO) and Brownian motion. These strategies effectively balance solution diversity and convergence speed, resulting in improved optimization performance and prevention of local optima.

Subsequently, in their investigation (Yu, Li, Li, Chen, Tang and Geng, 2024), scientists exceeded previous endeavors by assembling a comprehensive brain tumor dataset from multiple centers. This dataset encompasses diverse clinical brain tumor data, comprising segmentation and classification annotations. In order to enhance the precision of brain tumor segmentation, they introduced a novel approach named HSA-Net. This approach utilizes the Shared Weight Dilated Convolution module (SWDC) and the Hybrid Dense Dilated Convolution module (HDense) to effectively capture multi-scale information while minimizing the number of parameters.

In a recent research conducted by Ranjbarzadeh, Zarbakhsh, Caputo, Tirkolaee and Bendechache (2024), a novel brain tumor segmentation framework was introduced, which involves the utilization of four MRI sequence images. This framework incorporates an optimized CNN, where all weight and bias values are fine-tuned using an Improved Chimp Optimization Algorithm (IChOA). The initial step involves normalizing the four input images to detect potential tumor regions. Subsequently, the IChOA is applied to identify the most relevant features through a Support Vector Machine (SVM) classifier. Finally, these selected features are fed into the optimized CNN model to classify each object for brain tumor segmentation. The IChOA plays a crucial role in feature selection and hyperparameter optimization within the CNN model. The next section provides details of the proposed methodology in the current study.

3. Methodology

The proposed method in this research comprises four stages: preprocessing, image segmentation, feature analysis (including feature extraction and feature selection), and classification of identified tumors into high-grade and low-grade glioma categories. The preprocessing stage includes techniques to enhance the quality of MRI images. In the segmentation of MRI images, the K-means clustering algorithm is employed. To determine the centers of the clusters, innovative algorithms such as coyote optimization and coronavirus optimization are utilized. Various methods are also employed for feature extraction in this research, such as:

- Extract texture features using local binary patterns
- Training convolutional neural networks from scratch and extracting deep features
- Using the pre-trained VGG16 network for feature extraction
- Combination of LBP features and deep features obtained from the trained network and VGG16

Classification methods that will be used in this research include support vector machines, KNN, ECOC, decision tree, hybrid methods, and the Softmax classifier. More details of the proposed research method will be provided in the rest of this section. Figure 5 provides an overview of the proposed method.

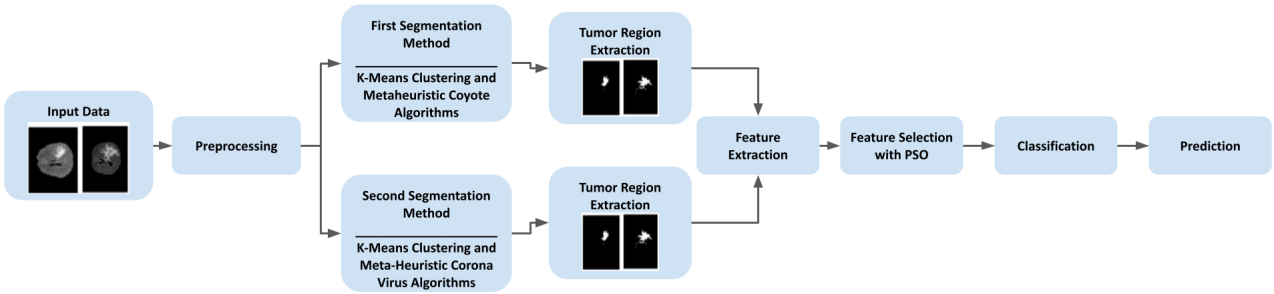

Figure 5: The overall architecture of the proposed methodology.

3.1. Input preprocessing

The MRI images used in this study are in .nii format, which were initially loaded using the NIFTI toolbox. In the first stage of preprocessing, the images are converted to grayscale. Selected slices of images are then saved in PNG format for subsequent preprocessing steps. In some sections of the MRI images, the motor area is unclear. Therefore, sections of the MRI that provide a better representation of the tumor area are selected, and the MRI images are mapped to a two-dimensional space. The Regionprops function in binary images calculates bounding component properties such as bounding box, pixel indices, and area. After applying this function, the potential background of the MRI images is removed. This operation is performed with the aim of extracting the main areas of the brain and other possible areas in the images.

3.2. MRI image segmentation

In this research, the K-means clustering algorithm is utilized for image segmentation. Identifying the most suitable center for K-means clustering is one of the most critical criteria for successful clustering. Since finding the optimal clustering centers is an NP-hard problem, meta-heuristic algorithms can serve as a guiding strategy.

3.2.1. K-means optimization with Coyote

In this research, one of the meta-heuristic algorithms employed is coyote optimization, as demonstrated in Algorithm 1. This process begins by calling the segmentation function, which has three arguments: the input image, the number of clusters, and the optimization method for cluster center points. After determining the optimal cluster center points, the image pixels are clustered using the K-Means algorithm. Ten percent of the initial cluster centers are generated using the K-Means algorithm, while the remainder are randomly generated. Following the random assignment of coyotes to different packs, the cost function for the initial social position of the population is examined. This cost function, which includes intra-cluster distance, inter-cluster distance, and entropy, aims to decrease the intra-cluster distance and increase the inter-cluster distance.

Intra-cluster distance measures the degree of similarity within a cluster. This criterion is obtained in the segmentation function by calculating the distance of each pixel within a cluster to the center of that cluster, and then averaging the total distances. Inter-cluster distance measures the dissimilarity between different clusters. This criterion is calculated by determining the centers of different clusters and calculating the difference between the highest light intensity value of one cluster and the lowest value of another cluster. Entropy is also used to evaluate the homogeneity of data within a specific cluster. The higher the entropy, the more heterogeneous the dataset, and vice versa. In the cost function, the objective is to minimize this criterion.

During the implementation of the COA algorithm, the fitness of each coyote in the population is evaluated multiple times until it reaches the maximum allowed number for the algorithm, which is 200 in this research. Finally, after obtaining the optimal cluster centers with the Coyote algorithm, the images are segmented using the K-Means method. In this method, the Euclidean distance is used to calculate the distance of each pixel to the center, and the pixels closest to each cluster center are determined for segmentation.

Algorithm 1 Coyote Optimization

Define search domain D, upper bound ub, and lower bound lb of decision variable j
Define number of packages N_p and number of coyotes N_c in each package
Set initial social conditions for each coyote $soc_c^{p,t} = (x_1, x_2, ..., x_D)$
Initialize coyote population randomly: $soc_c^{p,t} = lb_j + r_j \cdot (ub_j - lb_j)$ where r_j is random number in [0, 1]
Evaluate compatibility of coyotes: $Fit_c^{p,t} = f(soc_c^{p,t})$
Calculate alpha coyote for each pack
Calculate social orientation of each pack of coyotes
Update social status of each coyote within the pack and evaluate new status based on fitness function
Handle birth and death of coyotes
Move coyotes between packs with probability $P_e = 0.005 \cdot N_c^2$

3.2.2. K-means optimization with Corona virus

In this method, 10% of the initial positions of the viruses are determined through the K-means clustering algorithm, and the rest of the population positions are randomly determined between the upper and lower limits of the search space. During the optimization process, the position corresponding to the best fit value is obtained. In other words, the optimal cluster centers are generated. We can then perform the steps related to segmentation and assign each pixel of the input image to the closest center using the distance criterion. Algorithm 2 provides further details of this optimization method.

3.2.3. Tumor region extraction

First, the brightness intensity value in the segmented image is obtained using the unique() function and sorted in descending order. The pixel index of the image for each brightness value is found with the find command. After that,

Algorithm 2 Corona Virus Optimization

```
Initialize population randomly and evaluate fitness for each solution
Sort solutions in ascending order based on fitness, consider the best solution as initial best
while termination criteria not met do
    Select parent using Roulette Wheel selection
    Apply frameshifting technique to produce viral proteins
    for each new viral protein S_K do
        if forward frameshifting technique then
            S_K(1) = rand(minVal, maxVal)
            S_K(2 : D) = P(1 : D − 1)
        else
            S_K(D) = rand(minVal, maxVal)
            S_K(1 : D − 1) = P(2 : D)
        end if
    end for
    Combine viral proteins to produce new solution through Uniform Crossover
    Apply random mutations with mutation rate MR
    Evaluate objective function for new solution
    Update population for next generation (retain solutions with highest fitness)
end while
```

areas with fewer than 50 pixels in the image are not considered. As the process continues, a binary mask is created in which the values of ROI pixels are set to 1 and the rest of the image pixels are set to zero. With the help of morphological operations and the imfill command, the empty holes in the binary mask are filled. To extract the ROI region, the binary mask is multiplied by the original pre-processed image, resulting in the extraction of the regions belonging to the tumor.

3.3. Feature extraction

After segmenting the images and extracting the tumor area, it is time to extract features from the ROI areas. At this stage, several methods are employed: local binary patterns, initial training of a convolutional neural network, extraction of deep features from the pre-trained VGG16 network, and the combination of each of these sets in pairs with each other, along with the PSO method.

3.3.1. Local Binary Pattern

In this research, two sets of features are extracted from LBP based on different radius configurations and interpolation methods. In both sets of feature extraction with LBP, the number of neighboring pixels is set to 8. However, in the first approach, the radius value is 1, and linear interpolation and histogram normalization are used. In contrast, in the second approach, the nearest interpolation method is used without histogram normalization. Finally, these two sets of features obtained from the two approaches are combined into a feature vector.

3.3.2. Training a neural network from scratch

The purpose of designing a convolutional neural network is to extract deep features from tumor regions. For feature extraction using neural networks, the size of the tumor region images is standardized to 100*100. The images are also grayscale and have only one channel. The designed neural network is of the CNN type and includes two convolution layers, two size reduction layers, and two fully connected layers. The last layer is a SoftMax layer, which is defined for classification. Figure 6 illustrates the architecture of the designed neural network.

3.3.3. VGG16 Features

In this research, the VGG16 model is first initialized in MATLAB without pre-trained weights, meaning that the VGG16 layers are randomly initialized instead of using pre-trained weights from ImageNet. In the next step, the fully connected layer of the network is replaced with a new fully connected layer with two output classes. In addition to the output layer, the input layer is also set with 100*100 single-channel data. In a transfer learning approach called

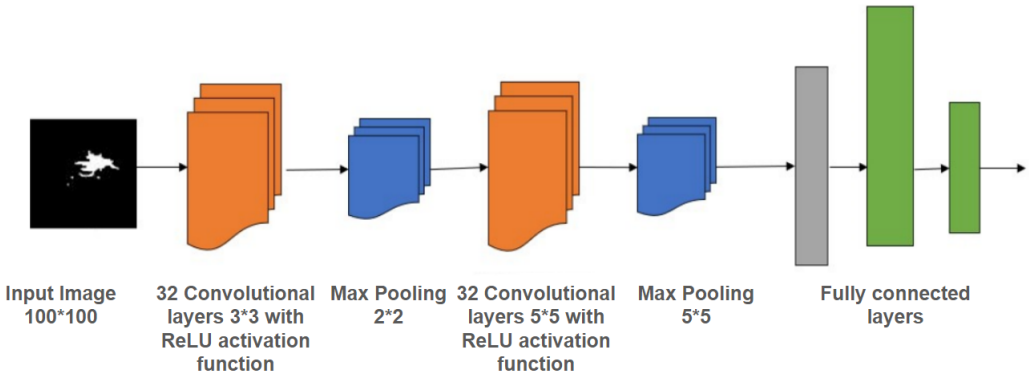

Figure 6: The neural network architecture of the proposed model.

fine-tuning, the entire network, including all the original layers of the VGG16 network and the modified layers, updates their weights during training.

After extracting features from each of the LBP, neural networks, and VGG16 methods, the set of features extracted from LBP is combined with each of the neural network and VGG16 features. These combined features are then selected as input to the PSO algorithm.

3.4. Feature selection

Dimensionality reduction methods transfer data from a high-dimensional feature space to a low-dimensional feature space, while preserving meaningful features. On the other hand, the feature selection approach is to select a small subset of the main features by removing irrelevant features. Feature selection leads to better learning of the model, higher accuracy, lower computational cost, and better interpretability of the model.

In this research, the particle optimization algorithm is used to select features. First, the parameters related to the PSO method, such as the number of decision variables, upper and lower bounds of the search space variable, inertia weight, individual and general learning coefficients, and a population of particles with random positions and zero velocity, are initialized. The best positions and personal and overall costs are determined based on initial evaluations. Then, the algorithm enters the main PSO loop for a certain number of iterations. For each iteration, particle velocities are updated based on personal and global best positions, with random components, and particle positions change using their velocities. Finally, the particle costs are updated using the evaluation fitness function, and the best positions and personal costs are updated if better solutions are found.

In this research, the correlation criterion, which is a statistical relationship between two random variables or two data sets, has been used for the objective function. In the given fit function, feature correlation refers to the relationships between the selected features and the output labels (the target variable). In this function, the correlation between pairs of features and the average correlation for selected features in a specific feature set are first calculated. A high level of this criterion indicates a high correlation of the features within the subgroup. Then, the correlation between each selected feature and the output labels is evaluated. In this research, the fitting function is evaluated through Equation 1 for each feature subset selected by the particle optimization algorithm. In this Equation, $_s$ is a subset containing N features, r_{cf} is the average feature-class correlation, and r_{ff} is the average feature-feature correlation. This Equation is also known as Pearsons correlation.

$$Fitness_s = \frac{N r_{cf}}{\sqrt{N + N(N-1) r_{ff}}} \tag{1}$$

3.5. Classification

Classification is a supervised learning approach that is used to analyze a given data set to build a model that divides the data into an arbitrary and specified number of classes. On the other hand, cross-validation is a technique used to evaluate the performance of a machine learning model. In cross-validation, the data are divided into two groups: training data and test data. By using this technique, a stronger estimate of the models performance is obtained on new and unseen data, and the overfitting of the model is avoided. The K-Fold evaluation method is used in this research. In this method, the data set is first separated into k subsets of equal volume. Then, in each iteration of the evaluation, k-1 subsets are considered as the training data set and one subset as the test set. In each iteration, the training and test subsets are changed, and at the end, the average error of the model in different iterations is calculated as the overall error. After dividing the data in this way, it is time to classify with different machine learning algorithms: Support Vector Machine, SoftMax Classification, Decision Tree, K-Nearest Neighbor, Error Correcting Output Codes (ECOC), and Ensemble Model (ECOC+SVM+SoftMax).

In the ECOC Method, we train an error-correcting coding classifier using binary support vector machines as learners. The ECOC classifier is trained on the training data and their corresponding class labels using the fitcecoc function, and it predicts the labels of the test data. Furthermore, Hybrid methods combine the forecasts of several models so that the final forecast produced is more accurate than any other model. This approach can improve the accuracy and robustness of classification models, enhance the ability to manage and process noisy data, and prevent overfitting. In this research, the final prediction of classification is made based on majority votes. The results obtained from the proposed methods will be discussed in the next section.

4. Experimental setting

The aim of this research is to detect glioma tumors from MRI images and classify them based on image processing, machine learning, and deep learning methods. In this section, first, the data set used in the research will be explained in detail along with other experimental settings.

4.1. Dataset

In this research, the Brats2018 dataset was used, which was collected from three different sources, including Brats2013, CBICA (Center for Biomedicine Image Computing), and ICIA (International Congress of Immunology and Allergy). The images in this dataset are divided into two categories: high-grade glioma (HGG) and low-grade glioma (LGG). The HGG group includes 20 cases from the Brats2013 group, 88 cases from CBICA, and 102 cases from ICIA. For the LGG images, 10 are from the Brats2013 group, and 65 are from ICIA. Each of the MRI images in this dataset includes four scanning methods: T2, T1GD, T1, and T2-FLAIR. All the images are recorded in a common pattern with the size of 150*240*240 voxels and have been sampled with an isotropic resolution of one millimeter. In this research, the T2-FLAIR image group was used due to its better quality. This dataset is publicly available on the Kaggle site. Examples of high-grade and low-grade glioma tumor data are shown in Figures 7 and 8, respectively.

4.2. Evaluation metrics

Various evaluation criteria for classifiers, including sensitivity (the rate of correct positive responses), specificity or detectability (the rate of correct negative responses), precision, accuracy, and the Dice coefficient, are used to evaluate the similarity between the extracted tumor and the corresponding true mask. To measure the two criteria of sensitivity and specificity, we require four parameters as below:

- True Positive: It is correctly identified.
- False positive: wrongly identified.
- True Negative: It has been correctly rejected.
- False Negative: It is wrongly rejected.

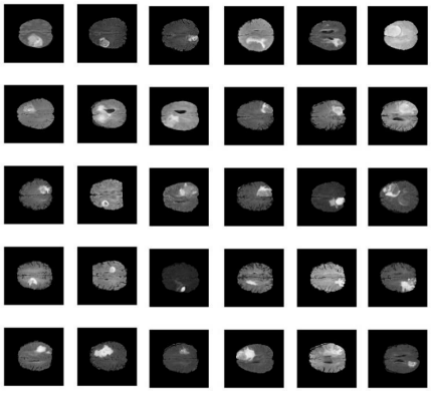

Figure 7: Sample images of high-grade glioma tumors.

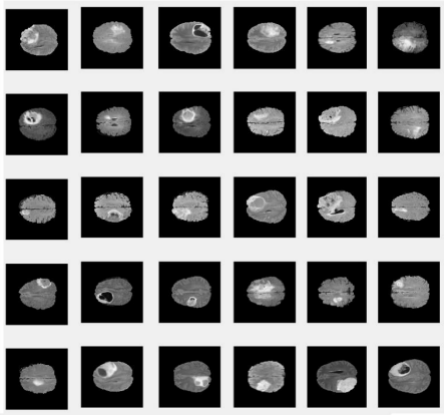

Figure 8: Sample images of low-grade glioma tumors.

Sensitivity: Sensitivity, also known as the True Positive Rate (TPR), measures a classifiers ability to correctly identify positive samples from the total number of actual positive samples. This measure is obtained by dividing the number of correctly predicted positive samples by the total number of actual positive samples. Sensitivity is calculated using Equation 2.

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

Specificity: Specificity, also known as the True Negative Rate (TNR), is a common criterion that is often calculated alongside sensitivity. This measure is the ratio of correctly predicted negative samples to the total number of actual negative samples. Specificity is calculated using Equation 3.

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{3}$$

Precision: This metric expresses the ratio of the number of true positive predictions to the total number of positive predictions made for examples of a class. High precision indicates a small number of false positives, or data that are wrongly classified as belonging to a particular class. This criterion is defined by Equation 4.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

Table 1
Parameters related to the first segmentation method.

Parameter	Value
Number of Clusters	4
Population	150
The number of boxes	10
Number of Iterations	200

Table 2
Parameters related to the second segmentation method.

Parameter	Value
Number of Clusters	4
Number of Viruses	50
Number of subproteins	2
Number of Iterations	30

Accuracy: Accuracy is a measure that is calculated as the number of correct predictions made by the classifier, divided by the total number of predictions made by the same classifier. While this criterion provides a general overview of the models performance, it does not offer detailed insights into the models efficiency. Accuracy is expressed by Equation 5.

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{TP} + \text{FN} + \text{FP}} \tag{5}$$

DICE Equation: In this validation method, the tumor area extracted by two segmentation methods is compared to the tumor area manually extracted by experts. In essence, this criterion assesses the similarity between the results of the segmentation methods and the actual mask. The Dice coefficient, which quantifies this similarity, is defined by Equation 6.

$$\text{Dice} = \frac{2(\text{I}_{\text{gt}} \cap \text{I}_{\text{ROI}})}{(\text{I}_{\text{gt}} \cup \text{I}_{\text{ROI}})} \tag{6}$$

4.3. Implementation

In this research, we used the MATLAB 2022 programming environment for coding. The MRI images used in this study are 3D and in the NIFTI file format. The NIFTI format is commonly used to store and share neuroimaging data, including MRI scans, Diffusion Tensor Imaging (DTI), and other types of brain imaging data. We used the NIFTI toolbox to import these images into the MATLAB environment. Additionally, we utilized machine learning and deep learning toolkits to train neural networks. The system used for this study includes a 7-core central processor, a graphics card with 4 GB of memory, and 12 GB of RAM.

4.4. Segmentation and neural network parameters

Tables 1 and 2 show the parameters of the first and second segmentation methods, respectively. In one of the feature extraction methods, a convolutional neural network model was used. Various parameters were set for its training, including the optimization algorithm, learning rate, and the number of repetitions of the training process. These were set to Adam, 0.001, and 30, respectively. The same values were also used for the pre-trained VGG model.

The particle swarm optimization algorithm, like all optimization algorithms, requires the setting of initial parameters. These include the number of particles and the number of iterations of the algorithm. In addition to these parameters, the learning rate and inertia weight are also set before running the algorithm. This is done to balance the exploration and exploitation process, as shown in Table 3.

Table 3
Parameters related to PSO algorithm.

Parameter	Value
Number of Seeds	50
Number of Iterations	250
C1 Coefficient	1.5
C2 Coefficient	2
Inertia Weight (w)	1

5. Results

This section delves into the details regarding the results obtained through the first and second segmentation approaches for tumor detection from brain images. This section is presented through three subsequent subsections. The last subsection, provides a comparative analysis between the proposed methodology and state of the art.

5.1. First segmentation method

Figure 9 displays a sample ROI extracted using the first segmentation method, alongside the actual mask corresponding to the tumor area. The input slices were randomly selected from the dataset. After calculating the Dice coefficient, we examined the percentage similarity between these extracted tumor samples and the actual masks of the ROI. The results of this examination are presented in Table 4.

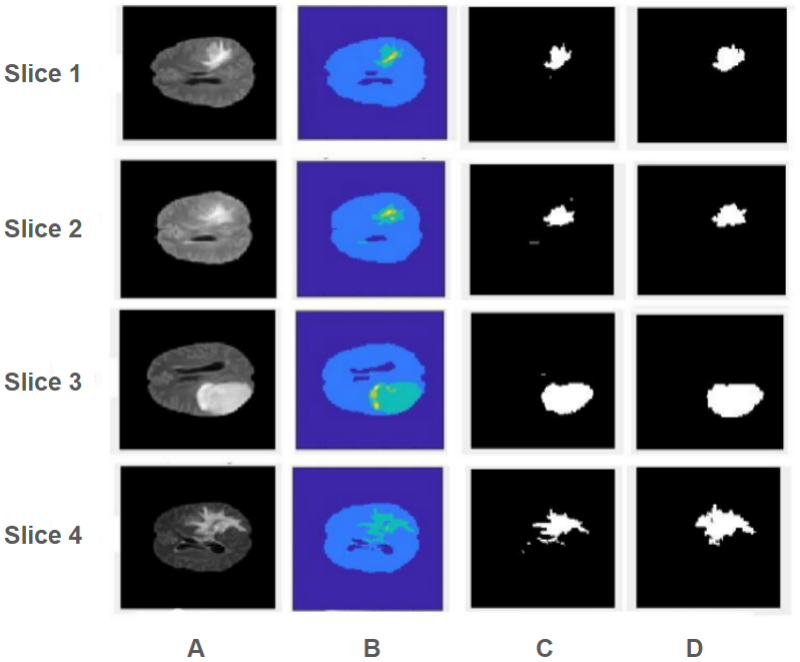

Figure 9: ROI extracted from the first segmentation method: (a) input image b) segmented image c) tumor area extracted from the first segmentation method d) true mask.

In this section, we examine the results obtained from two segmentation methods using various feature extraction techniques. These techniques are detailed in Tables 5 to 7 for individual methods and in Tables 8 and 9 for ensemble feature extraction methods. Each table provides details for 5-fold and 10-fold cross-validation across six classifiers. The evaluation metrics include Accuracy, Sensitivity, Specificity, and Precision.

Table 4
Dice coefficient results in the first segmentation method.

Slice Number	Dice Coefficient (%)
1	90.40
2	91.58
3	94.86
4	86.65
Average	90.87

Table 5
Final results of the LBP feature extraction method in the first method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	73.20	75.68	70.61	72.83
Decision Tree	74.80	75.29	74.28	75.29
KNN	71	74.11	67.75	70.52
ECOC	**80**	**79.29**	**80.81**	**81.12**
Ensemble Model	74.40	76.86	71.83	73.96
SoftMax	73.60	75.29	71.83	73.56
Evaluation Metrics (10-fold)				
SVM	71	74.50	67.34	70.37
Decision Tree	73.80	73.72	73.87	74.60
KNN	74	75.29	72.65	74.13
ECOC	**76**	**76.87**	**75.10**	**76.26**
Ensemble Model	72.60	73.33	71.83	73.04
SoftMax	71.40	74.90	67.75	70.74

We begin with Table 5, which presents the results of the LBP feature extraction for the first segmentation method. In the 5-fold cross-validation, the ECOC classifier outperforms the others across all four evaluation metrics. Conversely, the KNN classifier underperforms compared to the other five classifiers, achieving the lowest values for Specificity and Accuracy at 67.75% and 71%, respectively.

When examining the classifiers performance through 10-fold cross-validation, there is a slight decrease in the metrics. However, the ECOC classifier still maintains its superiority, achieving the highest value for Sensitivity at 76.87%. Overall, according to Table 5, the LBP feature extraction method yields better results for classifiers through 5-fold cross-validation.

Nevertheless, the ANN feature extraction yielded varied results for the ECOC and KNN classifiers. According to Table 6, for 5-fold cross-validation, both KNN and ECOC achieved the highest Accuracy value of 97.60%. However, KNN outperformed ECOC in terms of Sensitivity, achieving a value of 98.89%. ECOC, on the other hand, led in terms of Specificity and Precision, with values of 98.73% and 96.91%, respectively. In contrast, for 10-fold cross-validation, SoftMax surpassed ECOC with the highest Sensitivity value of approximately 99%. Despite this, ECOC maintained its superiority over other classifiers in terms of Accuracy, Specificity, and Precision, with values of 98.40% and slightly above.

Table 7 provides further details on the performance of VGG16 feature extraction for both 5-fold and 10-fold cross-validation. A general observation of this table reveals minor differences between the ECOC and SoftMax classifiers in terms of Accuracy, as both achieved a value of 99%. This trend is also observed for SVM, ECOC, and SoftMax, all of which achieved similar Sensitivity values of 98.82%.

Table 6
Final results of the ANN feature extraction method in the first method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	97.40	98.43	96.32	96.53
Decision Tree	96.20	96.86	95.51	95.73
KNN	**97.60**	**98.89**	96.32	96.55
ECOC	**97.60**	98.43	**98.73**	**96.91**
Ensemble Model	97.40	97.14	96.32	96.53
SoftMax	97	97.21	94.69	95.11
Evaluation Metrics (10-fold)				
SVM	98.20	98.43	97.95	98.04
Decision Tree	95.40	96.47	94.28	94.61
KNN	97.80	98.82	96.73	96.92
ECOC	**98.40**	98.43	**98.36**	**98.43**
Ensemble Model	98.20	98.14	97.95	98.04
SoftMax	97.80	**99.21**	96.32	96.56

Table 7
Final results of the VGG16 feature extraction method in the first method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	98.20	**98.82**	97.55	97.67
Decision Tree	97.20	97.64	96.73	96.88
KNN	97.40	98.03	96.73	96.89
ECOC	**99**	**98.82**	99.18	99.21
Ensemble Model	98.60	98.03	97.55	99.12
SoftMax	**99**	**98.82**	**99.73**	**99.89**
Evaluation Metrics (10-fold)				
SVM	98.40	**98.82**	97.95	98.05
Decision Tree	98.80	98.43	99.18	99.20
KNN	97.60	98.43	96.73	96.91
ECOC	98.80	98.14	99.18	98.20
Ensemble Model	98.80	**98.82**	98.77	98.43
SoftMax	**99**	**98.82**	**99.18**	**99.20**

Conversely, only one classifier achieved the highest Precision of 99.89%, with a negligible difference compared to that obtained by ECOC. Moreover, in 10-fold cross-validation, the highest Accuracy and Precision were obtained by the SoftMax classifier. However, for Sensitivity, the highest value was similarly achieved by SVM, Ensemble Model, and SoftMax at 98.82%. This marks the first time that the Ensemble Model ranked among the top in terms of its performance across evaluation metrics. Additionally, the Decision Tree classifier, with a Specificity value of 99.18%, performed on par with SoftMax.

Tables 8 and 9 present the performance of ensemble feature extraction methods, such as ANN+LBP and VGG16+LBP, for the first segmentation method. According to Table 8, SoftMax leads with the highest obtained values of over 97.5% across all metrics, with the exception of the Ensemble Model, which demonstrated similar performance to SoftMax in terms of Specificity. This trend continues for 10-fold cross-validation, but there is a significant increase in the Accuracy and Specificity values for the Ensemble Model and ECOC, respectively.

Based on the data in Table 9, the best Accuracy for VGG16+LBP feature extraction is achieved by the ECOC and Decision Tree classifiers, both with a value of 98%. However, for Sensitivity, KNN stands out with a value of 98.82%. For the Ensemble Model, the highest rates are achieved for Specificity and Precision, at 98.36% and

Table 8
Final results of the ANN+LBP feature extraction method in the first method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	97	97.25	96.73	96.87
Decision Tree	96.80	96.86	96.47	96.86
KNN	95	94.50	95.51	95.63
ECOC	97.60	97.64	**97.55**	97.64
Ensemble Model	97.40	97.71	97.14	97.26
SoftMax	**97.80**	**98.03**	**97.55**	**97.65**
Evaluation Metrics (10-fold)				
SVM	97.40	97.64	97.14	97.26
Decision Tree	96	96.47	95.51	95.71
KNN	96.60	96.86	96.32	96.48
ECOC	97.80	98.03	**97.55**	97.65
Ensemble Model	**98**	98.14	**97.55**	97.65
SoftMax	**98**	**98.43**	**97.55**	**98.66**

Table 9
Final results of the VGG16+LBP feature extraction method in the first method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	97.20	97.64	96.73	96.88
Decision Tree	98	98.43	97.55	97.66
KNN	97.40	98.82	95.91	96.18
ECOC	**98**	**97.64**	**98.36**	**98.41**
Ensemble Model	97.40	98.03	96.73	96.89
SoftMax	97.60	98.03	97.14	97.27
Evaluation Metrics (10-fold)				
SVM	93	97.25	88.57	89.85
Decision Tree	99	99.21	98.77	98.82
KNN	**99**	**98.82**	**99.18**	**99.21**
ECOC	98.60	98.35	98.36	98.43
Ensemble Model	98.60	99.21	98.95	98.06
SoftMax	98.20	97.64	98.77	98.80

98.41% respectively, in 5-fold cross-validation. In 10-fold cross-validation, the Decision Tree and KNN classifiers show comparable performances in terms of Accuracy, both with a value of 99%. For Sensitivity, the highest value is jointly achieved by the Decision Tree and the Ensemble Model, at 99.21%. Moreover, KNN achieves the highest values of Specificity and Precision, at 99.18% and 99.21%, respectively.

5.2. Second segmentation approach

In this section, we present the results of the proposed second method for brain tumor image segmentation. The classification results of the feature extraction methods and the classification using the second method are reported in the subsequent tables. Figure 10 illustrates the tumor regions extracted using this method. The calculated Dice coefficients, which are also presented in Table 10, indicate better performance compared to the first segmentation method on similar images.

Table 11 shows that the Dice coefficients for the second segmentation method have increased to an average of 92.36%, compared to the value of 90.87% in the first segmentation method. Table 12 presents the performance of the six suggested classifiers in this study for the LBP feature extraction in the second segmentation method. For the 5-fold

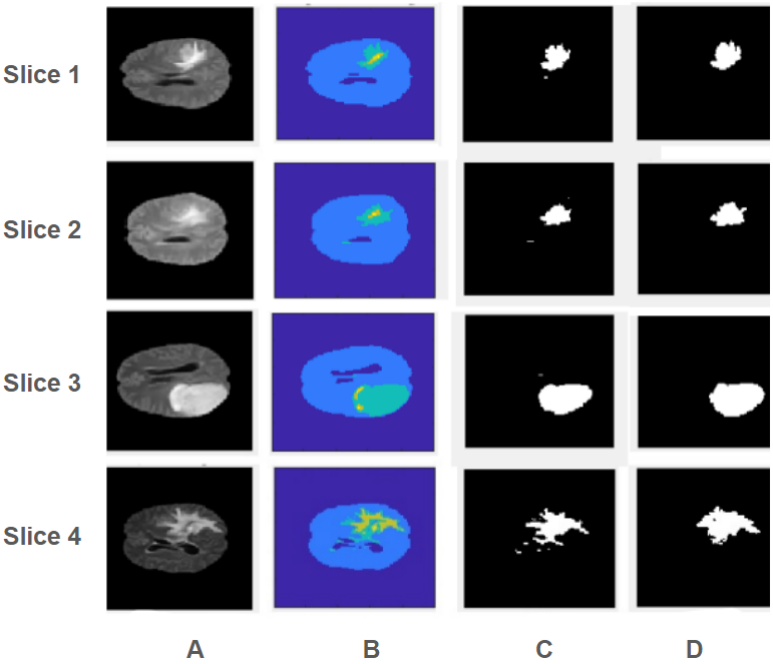

Figure 10: ROI extracted from the second segmentation method: (a) input image b) segmented image c) tumor area extracted from the first segmentation method d) true mask.

Table 10
Dice coefficient results in the second segmentation method.

Slice Number	Dice Coefficient (%)
1	91.56
2	91.73
3	95.54
4	90.64
Average	92.36

cross-validation, the KNN classifier registers the top rates with Accuracy at 82.80%, Sensitivity at 83.52%, Specificity at 82.04%, and Precision at 82.87%. The same top performance is repeated for KNN in 10-fold cross-validation. Conversely, the Decision Tree classifier achieves the lowest performance, with Accuracy at 69% and Sensitivity at 68.62%, representing the lowest achieved rates.

Tables 12 and 13 present the performance of the ANN and VGG16 feature extraction methods for the second segmentation method. According to Table 12, SVM is ranked as the best-performing classifier for both 5-fold and 10-fold cross-validation, achieving the highest Sensitivity value of 100%. On the other hand, Table 13 shows that SoftMax outperforms other classifiers across all evaluation metrics for both 5-fold and 10-fold cross-validation, achieving the highest Accuracy value of 99.80%.

Following these results, the combination of ANN and LBP for feature extraction yielded the best results for the ECOC classifier, as illustrated in Table 14. The highest values achieved were 99.60% for the metrics of Accuracy, Sensitivity, and Precision in 5-fold cross-validation, and 100% for Sensitivity in 10-fold cross-validation. Table 15 ranks KNN as the highest-performing classifier when applied to VGG16 and LBP feature extraction. For both 5-fold and 10-fold cross-validation, the highest rates were obtained for Precision, with values of 98.81% and 99.21%

Table 11
Final results of the LBP feature extraction method in the second method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	74.40	75.68	73.06	74.51
Decision Tree	72.20	71.76	72.65	73.20
KNN	**82.80**	**83.52**	**82.04**	**82.87**
ECOC	72.20	72.94	71.94	72.65
Ensemble Model	75	76.86	73.06	74.80
SoftMax	74.40	76.47	72.24	74.14
Evaluation Metrics (10-fold)				
SVM	72	72.94	71.02	72.37
Decision Tree	69	68.62	69.38	70
KNN	**84.60**	**84.31**	**84.89**	**85.31**
ECOC	73.80	74.11	73.46	74.40
Ensemble Model	73.60	74.50	72.65	73.93
SoftMax	75	76.07	73.87	75.93

Table 12
Final results of the ANN feature extraction method in the second method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	**99.80**	**100**	**99.59**	**99.60**
Decision Tree	98.40	98.82	97.14	97.32
KNN	99.60	100	99.18	99.22
ECOC	99.60	99.87	99.59	99.60
Ensemble Model	99.60	99.60	99.59	99.60
SoftMax	99.20	98.82	99.18	99.47
Evaluation Metrics (10-fold)				
SVM	**99.80**	**100**	**99.59**	**99.60**
Decision Tree	99	98.82	99.18	99.21
KNN	99.40	100	98.77	98.83
ECOC	99.60	99.18	98.82	99.60
Ensemble Model	99.60	100	98.18	98.83
SoftMax	99.20	98.82	99.59	99.47

respectively, slightly surpassing the values for Accuracy and Specificity.

5.3. Comparative analysis

In recent research, various methods have been employed to diagnose brain tumors, including glioma, and to grade them. In this section, we present the results of the second segmentation method and the feature extraction method of the trained network, which have demonstrated superior results compared to other methods. These results are compared with some prominent methods in terms of performance.

Research (Sharif et al., 2020) pertains to the diagnosis of glioma tumors, and the PSO algorithm was used for image segmentation. Reference (Amin et al., 2020) employs the LSTM network to classify this type of disease. Similar to these works, the authors of (Rehman et al., 2021) have also used the dataset utilized in this thesis to identify the tumor area and grade it, achieving an accuracy of 92.67%. In reference (Daimary et al., 2020), a combination of CNN models was used for image segmentation, achieving a final accuracy of 99.11%. Table 16 presents a comparison of the results of the best method in this research with the aforementioned references.

Table 13
Final results of the VGG16 feature extraction method in the second method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	99.20	99.21	98.87	99.21
Decision Tree	98.80	98.82	97.78	98.82
KNN	98.40	98.43	98.36	98.43
ECOC	99	98.82	99.18	99.21
Ensemble Model	99.20	99.21	98.73	99.21
SoftMax	**99.20**	**99.57**	**99.18**	**99.57**
Evaluation Metrics (10-fold)				
SVM	99.20	99.21	99.18	99.21
Decision Tree	98	97.25	98.77	98.80
KNN	98.80	98.82	98.55	98.82
ECOC	99.60	99.60	99.59	99.60
Ensemble Model	99.60	99.40	99.59	99.24
SoftMax	**99.80**	**99.78**	**99.70**	**99.60**

Table 14
Final results of the ANN+LBP feature extraction method in the second method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	98.80	98.82	98.77	98.82
Decision Tree	99.40	99.60	99.18	99.21
KNN	99	99.35	98.36	98.44
ECOC	**99.60**	**99.60**	**99.59**	**99.60**
Ensemble Model	99.40	99.21	99.18	99.60
SoftMax	99.20	99.57	99.37	99.21
Evaluation Metrics (10-fold)				
SVM	99	99.21	98.77	98.82
Decision Tree	98.40	98.82	97.95	98.05
KNN	99.20	98.82	98.36	98.45
ECOC	**99.60**	**100**	**99.18**	**99.22**
Ensemble Model	99.40	99.60	98.18	99.21
SoftMax	99.40	99.60	98.18	99.21

6. Conclusion

In general, tumor identification and the diagnosis of the degree of this disease in CAD systems can be divided into four stages: pre-processing of images, segmentation, feature engineering (which includes feature extraction and feature selection), and the design of different classifiers to categorize tumors. In addition to image preprocessing, correct segmentation and accurate extraction of the tumor area can effectively improve the final result of the system.

Feature extraction is also very important in identifying the high and low grades of the identified tumor. In this research, we investigated the methods of pre-processing, segmentation, feature extraction, and feature selection for tumor identification and diagnosis of its grade. We used a combination of the K-means clustering algorithm and meta-heuristic algorithms of coyote optimization and the coronavirus optimization algorithm for image segmentation. The features extracted from the LBP method were combined with the features of convolutional neural networks and VGG16, and different classifiers were used to classify glioma brain tumors.

According to the conducted research, we hope to use adversarial generative networks to increase the data in future research. We also hope to be able to use other datasets, including CT scan images, to evaluate the performance of the proposed methods, and apply these methods to diagnose other types of tumors.

Table 15
Final results of the VGG16+LBP feature extraction method in the second method of segmentation.

Evaluation Metrics (5 Folds)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
SVM	97	97.64	96.32	96.51
Decision Tree	98	98.03	97.95	98.03
KNN	**98.80**	**98.43**	**98.77**	**98.81**
ECOC	98.60	99.60	97.95	98.06
Ensemble Model	98.60	98.82	98.36	98.43
SoftMax	98.40	98.03	98.77	98.81
Evaluation Metrics (10-fold)				
SVM	93	97.25	88.57	89.85
Decision Tree	99	99.21	98.77	98.82
KNN	**99**	**98.82**	**99.18**	**99.21**
ECOC	98.60	98.82	98.36	98.43
Ensemble Model	98.60	57.21	97.95	98.06
SoftMax	98.20	97.64	98.77	98.80

Table 16
Analysis and comparison of the results of this research with the results of related research.

Methodology	Dataset	Best Accuracy (%)
Segmentation of MRI Images by PSO Method, Tissue Feature Extraction with LBP and Deep Feature Composition Derived from encapsulated neural networks, using genetic algorithm to select feature and grading Glioma tumors using classifier ANN (Sharif et al., 2020)	Brats Challenge	99
Using LSTM network to classify glioma brain tumors (Amin et al., 2020)	Brats Challenge	98
Combining CNN models such as ResNet18, UNet and SegNet to segment MRI images (Daimary et al., 2020)	Brats Challenge	99.11
Using 3D CNN for segmentation and determination of tumor area, feature extraction using model Pre-trained VGG19 with transfer learning method (Rehman et al., 2021)	Brats Challenge	92.67
The proposed method (using optimal k-means) for optimization of cluster centers using Corona virus (meta-heuristic algorithm) for image segmentation, extracting deep features from the trained convolutional network, feature selection with PSO and use of SVM classifier to classify glioma tumors	Brats Challenge	99.80

CRediT authorship contribution statement

Hadis Rashno: Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Writing - Original Draft, Visualization. **Abdolah Chalechale**: Conceptualization, Formal Analysis, Writing - Original Draft, Validation, Writing - Review & Editing, Visualization, Supervision, Project Administration, Funding acquisition. **Faezeh Safari**: Writing - Original Draft, Investigation, Writing - Reviewing & Editing, Formal Analysis, Visualization, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors confirm that the access link to data analysed during this study are included in the paper.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: